SAP Cheat Sheet

S/4HANA 2025 FPS02 · Public Edition 2608 · CAP @sap/cds 10.2 · SAPUI5 1.154 · AI-generated summary — verify against help.sap.com

SAP GUI — TRANSACTION CODES

Command field

```
/nmb01   end task, open new transaction
/o SE80  open it in a new session
/o       list sessions
/i       close this session
/nend    confirm, then log off
/nex     log off at once, no dialog
```

Common codes

```
SU01 users        PFCG roles
SE80 Object Navigator
SM59 RFC destinations
STMS transports   SM37 job overview
SM12 lock entries ST22 ABAP dumps
/IWFND/MAINT_SERVICE  OData services
```

CLIENTS, LANDSCAPE, TRANSPORTS

Clients and systems

```
DEV -> QAS -> PRD   three-system group
T000    client control table
SCC4    client settings
MANDT   client key column
```

Transports

```
SE09 / SE01   request overview
STMS          Transport Management
S_TRANSPRT    authorization object
```

Other tools

Public Edition: Adaptation Transport Organizer. BTP content: SAP Cloud Transport Management. SAP Cloud ALM orchestrates, it does not move transports.

USERS, ROLES, AUTHORIZATIONS

ABAP system

```
SU01   create user, assign roles
PFCG   roles, menu, authorizations
SU53   last failed check
SUIM   users, roles, critical rights
ST01   authorization trace
```

Other models

Public Edition: business roles from business catalogs, narrowed by restriction types. BTP: role templates, role collections, then users or groups.

ARCHITECTURE AND VDM

Three tiers

Presentation: Fiori launchpad, SAP GUI (DIAG). Application: ABAP server with SAP Gateway. Database: SAP HANA, in-memory and column-oriented.

View prefixes

```
I_  interface view
C_  consumption view
P_  private view
R_ A_ E_  further prefixes
```

Note

Build only on released views (contracts C0, C1, C2).

ODATA AND GATEWAY

Query options

```
$metadata   service description
$filter=amount gt 50
$select  $orderby=ID  $top=5
$skip=10  $count=true  $expand=items
$batch      many requests, one trip
```

Gateway

```
SEGW                  build V2 services
/IWFND/MAINT_SERVICE  publish V2
/IWFND/V4_ADMIN       V4 service groups
x-csrf-token: fetch   before writes
```

INTEGRATION — RFC, BAPI, IDOC

ALE chain of settings

```
SM59   destination
WE21   port
WE20   partner profile
WE41 / WE42   process codes
WE02   display IDocs
WE19   test tool
```

Notes

BAPI: RFC-capable function module, synchronous via RFC, asynchronous via ALE. Commit with BAPI_TRANSACTION_COMMIT. ABAP Cloud: communication arrangements. Prefer OData and events for new interfaces.

BTP AND CAP COMMANDS

Project and local run

```
npm i -g @sap/cds-dk
cds init expenses --nodejs
cds watch
cds version
cds env get requires.auth.kind
```

Deploy

```
cds add hana,xsuaa,mta
cds up
mbt build -t gen --mtar mta.tar
cf deploy gen/mta.tar -f
cf login --sso
cf target
cds add kyma
```

Accounts

Global account, subaccounts (one region each), optional directories. Cloud Foundry: org and spaces. Kyma: clusters and namespaces.

SAPUI5 AND FIORI

UI5 CLI

```
npx ui5 serve --port 8127
npx ui5 build --dest ../dist
```

App parts

Component with manifest.json (models, libraries, routes), XML views, JavaScript controllers loaded with sap.ui.define, models, i18n files, routes.

Fiori elements annotations

```
@UI.LineItem         list report columns
@UI.SelectionFields  filter bar fields
@UI.HeaderInfo       object names, title
@UI.Facets           object page sections
@UI.FieldGroup       fields in a section
```

EXTENSIBILITY, DEPLOYMENT

Three extension kinds

Key user (low-code, in the app), developer (ABAP in the S/4HANA stack), side-by-side (on SAP BTP). Start with key user for small changes.

Clean core levels

A to D by the SAP objects used; ATC shows no finding, information, warning, error.

Deployment options

Public Edition: multi-tenant, upgrades February and August. Private Edition: single-tenant, RISE with SAP. On premise: customer operated. Paths: new implementation, system conversion, selective data transition.